

MATH340Fa22_Midterm2

Monday, November 7, 2022 9:45 AM



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Math 340: Elementary Matrix and Linear Algebra

MIDTERM TWO

Monday, November 14, 2022

Please circle your instructor and your Discussion Session (start time given) in the table below.

Dr. Skylar Grey	Dr. Boya Wen	Dr. Ziquan Yang
Zaidan Wu, T 3:30	Alex Pahlow, M 12:05	Connor Simpson, T 7:45
Zaidan Wu, T 4:35	Ugur Cin, M 12:05	Asvin Gothandaraman, T 8:50
Lauren Neurdorf, R 3:30	Ugur Cin, W 12:05	Connor Simpson, R 7:45
Hyun Jong Kim, R 9:55	Pubo Huang, W 1:20	Asvin Gothandaraman, R 9:55
Hyun Jong Kim, T 9:55	Woojin Kim, M 3:30	
Hyun Jong Kim, R 8:50	Woojin Kim, W 3:30	
	Ugur Cin, W 4:35	
	Woojin Kim, M 2:25	

Dr. Kaitlyn Phillipson	Dr. Zane Li
Emma Thomas, R 11:00	Jacob Denson, M 1:20
Emma Thomas, R 12:05	Jacob Denson, M 2:25
Emma Thomas, R 1:20	Jacob Denson, W 2:25

Name: **Key** Wisc email: _____

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: _____

READ THE FOLLOWING INFORMATION. **DO NOT BEGIN THIS EXAM UNTIL SIGNED TO DO SO.**

- This is a 90-minute exam. It consists of nine problems for a total of 50 points; the exam is 7 sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process. **An incomplete exam will result in an automatic zero.**
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.
- Notation reminders: M_{mn} is the vector space of $m \times n$ real matrices, P_d is the vector space of polynomials of degree at most d , $\mathbf{0}$ denotes the zero vector in a vector space V . $\mathbb{R}^n$ uses the normal vector addition/scalar multiplication unless otherwise noted.

1. (5 points, 1 point each) Are each of the following true or false statements? Circle your answer. Justification is not necessary.

- a.) ☒ **T** ☐ **F** The kernel of $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x-y \\ 2x+y \end{bmatrix}$ only consists of the zero vector $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$.
- b.) ☐ **T** ☒ **F** If A is a 3×3 matrix with $\det(A) = 2$, then the dimension of the nullspace of A is at least 2.
- c.) ☒ **T** ☐ **F** A set of 5 polynomials in P_3 is always linearly dependent.
- d.) ☐ **T** ☒ **F** If S is a subset of a vector space V and $\mathbf{0}$ is in S , then S is linearly independent.
- e.) ☐ **T** ☒ **F** $\mathbb{R}^5$ and M_{32} are isomorphic vector spaces.

2. (5 points) Consider the vector space M_{22} with normal matrix addition and scalar multiplication. Consider the subset W of M_{22} of matrices of the form $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where $a + d = 0$, $b = 3c$. Is W a subspace of M_{22} ? Completely justify your answer.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in W \Rightarrow \begin{bmatrix} a & 3c \\ c & -a \end{bmatrix} \text{ is the form.}$$

$\downarrow$
 $d = -a$

$$\begin{bmatrix} a_1 & 3c_1 \\ c_1 & -a_1 \end{bmatrix} + \begin{bmatrix} a_2 & 3c_2 \\ c_2 & -a_2 \end{bmatrix} = \begin{bmatrix} a_1 + a_2 & 3(c_1 + c_2) \\ c_1 + c_2 & -(a_1 + a_2) \end{bmatrix} \checkmark$$

$$r \begin{bmatrix} a_1 & 3c_1 \\ c_1 & -a_1 \end{bmatrix} = \begin{bmatrix} ra_1 & 3(rc_1) \\ rc_1 & -(ra_1) \end{bmatrix} \checkmark$$

Nonempty: $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 3 \cdot 0 \\ 0 & -0 \end{bmatrix} \in W \checkmark$

So W is a subspace!

3. (6 points total) Let F be the subset of vectors in $\mathbb{R}^2$ where $F = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x+y=2 \right\}$ under the operations

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \oplus \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 - 2 \end{bmatrix}$$

$$c \odot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} cx \\ cy - 2c + 2 \end{bmatrix}$$

- a. (3 points) Does there exist a zero vector $\mathbf{0}$ in F such that $\mathbf{0} \oplus \mathbf{u} = \mathbf{u} = \mathbf{u} \oplus \mathbf{0}$ for all $\mathbf{u}$ in F ? Justify your answer completely.

↳ $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$. let's check! (need to check both)

$$\begin{bmatrix} x \\ y \end{bmatrix} \oplus \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} x+0 \\ y+2-2 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \checkmark$$

$$\begin{bmatrix} 0 \\ 2 \end{bmatrix} \oplus \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0+x \\ 2+y-2 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \checkmark$$

Yes! $\vec{0} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$.

- b. (3 points) Show that the following vector space property holds for F :

$$c \odot (\mathbf{u} \oplus \mathbf{v}) = (c \odot \mathbf{u}) \oplus (c \odot \mathbf{v})$$

(more room on next page, if needed)

$$c \odot \left(\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \oplus \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \right) = c \odot \left(\begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 - 2 \end{bmatrix} \right) = \begin{bmatrix} c(x_1 + x_2) \\ c(y_1 + y_2 - 2) - 2c + 2 \end{bmatrix}$$

Overflow room for Problem 3:

$$\begin{aligned} \begin{bmatrix} cx_1 + cx_2 \\ cy_1 - 2c + 2 + cy_2 - 2c + 2 - 2 \end{bmatrix} &= \begin{bmatrix} cx_1 \\ cy_1 - 2c + 2 \end{bmatrix} \oplus \begin{bmatrix} cx_2 \\ cy_2 - 2c + 2 \end{bmatrix} \\ &= \left(\text{co} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \right) \oplus \left(\text{co} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \right) \quad \square. \end{aligned}$$

4. (6 points) Suppose A is a 4×5 matrix with the following reduced row echelon form:

$$\begin{array}{cccccc} & x_1 & x_2 & x_3 & x_4 & x_5 \\ \begin{bmatrix} 1 & 0 & 0 & 2 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

- a.) (2 points) What is the rank of A ? Explain your answer in one sentence.

rank(A) = 3 b/c there's 3 leading ones/pivots.

- b.) (2 points) Find, if possible, a basis for the nullspace of A . If there is not enough information, explain.

$$x_4 = s$$

$$x_5 = t$$

$$x_1 = -2s$$

$$x_2 = s$$

$$x_3 = -s$$

$$s \begin{bmatrix} -2 \\ 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

basis:

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

- c.) (2 points) Find, if possible, a basis for the span of the column vectors of A . If there is not enough information, explain.

can't do it $\rightarrow$ we know the first 3 column vectors work, but we don't know what they were before the reduction.

Overflow room for Problem 4, if needed:

5. (5 points total) For the set $S = \left\{ \begin{bmatrix} c \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ c \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \right\}$:

a.) (4 points) Find all values of c for which the set S is linearly independent. Fully justify your answer.

$$\begin{aligned} \det \begin{bmatrix} c & 0 & 1 \\ -1 & c & 1 \\ 1 & 2 & 2 \end{bmatrix} &= c(2c-2) + 1(-1 \cdot 2 - c) \\ &= 2c^2 - 2c - c - 2 \\ &= 2c^2 - 3c - 2 \\ &= (2c+1)(c-2) = 0 \\ & \quad c = -1/2, 2. \end{aligned}$$

If $c \neq -1/2, 2$, then the set is linearly independent.

b.) (1 point) Using your answer to part (a), which values of c make S a basis for $\mathbb{R}^3$? Justify your answer.

Since $c \neq -1/2, 2$ makes the 3 vectors lin. indep,
this makes them a basis for $\mathbb{R}^3$ b/c
There are 3 vectors and $\dim(\mathbb{R}^3) = 3$.

6. (5 points) Determine if the following set S of 2×2 matrices is linearly dependent:

$$S = \left\{ \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 3 \\ -1 & 2 \end{bmatrix} \right\}$$

$$a \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} + b \begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix} + c \begin{bmatrix} 0 & 3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{cases} 2a + 2b = 0 \\ 3a + 3c = 0 \\ -a - b - c = 0 \\ 2a + b + 2c = 0 \end{cases} \rightarrow \left[\begin{array}{ccc|c} 2 & 2 & 0 & 0 \\ 3 & 0 & 3 & 0 \\ -1 & -1 & -1 & 0 \\ 2 & 1 & 2 & 0 \end{array} \right]$$

$$\xrightarrow{\frac{1}{2}r_1 \rightarrow r_1, \frac{1}{3}r_2 \rightarrow r_2, -1r_3 \rightarrow r_3} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 2 & 1 & 2 & 0 \end{array} \right] \xrightarrow{\begin{array}{l} -r_1 + r_2 \rightarrow r_2 \\ -r_1 + r_3 \rightarrow r_3 \\ -2r_1 + r_4 \rightarrow r_4 \end{array}} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{array} \right] \xrightarrow{-r_2 + r_4 \rightarrow r_4} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \begin{cases} a + b = 0 \rightarrow a = 0 \\ b - c = 0 \rightarrow b = 0 \\ \uparrow \\ \rightarrow c = 0 \end{cases} \text{ so this means they are linearly independent.}$$

7. (6 points total) Each of the following statements is **false**. Show each statement is false by providing explicit counterexamples.

a.) (2 points) The set of vectors $\begin{bmatrix} a \\ b \end{bmatrix}$ where $ab \geq 0$ is a subspace of $\mathbb{R}^2$.

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix} + \begin{bmatrix} -2 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$1 \cdot 3 \geq 0 \checkmark$ $(-2)(-1) \geq 0 \checkmark$ $\hookrightarrow (-1)(2) < 0 \times$

b.) (2 points) The function $L: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $L\left(\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}\right) = \begin{bmatrix} u_1^2 \\ 3u_2 \end{bmatrix}$ is a linear transformation.

$$L\left(\begin{bmatrix} 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 3 \\ 0 \end{bmatrix}\right) = L\left(\begin{bmatrix} 5 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 25 \\ 0 \end{bmatrix}$$

vs.

$$L\left(\begin{bmatrix} 2 \\ 0 \end{bmatrix}\right) + L\left(\begin{bmatrix} 3 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 4 \\ 0 \end{bmatrix} + \begin{bmatrix} 9 \\ 0 \end{bmatrix} = \begin{bmatrix} 13 \\ 0 \end{bmatrix}$$

not the same.

(can also do: $L(2\begin{bmatrix} 1 \\ 0 \end{bmatrix}) = L(\begin{bmatrix} 2 \\ 0 \end{bmatrix}) = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$ vs. $2L(\begin{bmatrix} 1 \\ 0 \end{bmatrix}) = 2\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ not the same)

c.) (2 points) Let V be the set of real numbers with the operation $a \oplus b = a - b$. $\oplus$ satisfies the vector space property $\mathbf{v} \oplus \mathbf{u} = \mathbf{u} \oplus \mathbf{v}$.

$$3 \oplus 1 = 3 - 1 = 2$$

$$1 \oplus 3 = 1 - 3 = -2$$

not equal.

8. (8 points total) Let $L: P_2 \rightarrow P_1$ be the function defined by

$$L(at^2 + bt + c) = (a+b)t + (a-c).$$

a.) (2 points) Show that L is a linear transformation.

$$L(p+q) =$$

$$\begin{aligned} L(a_1t^2 + b_1t + c_1 + a_2t^2 + b_2t + c_2) &= L((a_1+a_2)t^2 + (b_1+b_2)t + c_1+c_2) \\ &= (a_1+a_2+b_1+b_2)t + (a_1+a_2-(c_1+c_2)) \\ &= (a_1+b_1)t + (a_1-c_1) + (a_2+b_2)t + (a_2-c_2) \\ &= L(a_1t^2 + b_1t + c_1) + L(a_2t^2 + b_2t + c_2) \\ &= L(p) + L(q) \end{aligned}$$

$$L(rp) =$$

$$\begin{aligned} L(r(at^2 + bt + c)) &= L(rat^2 + rbt + rc) \\ &= (ra+rb)t + (ra-rc) = r((a+b)t + (a-c)) \\ &= rL(at^2 + bt + c) \\ &= rL(p) \end{aligned}$$

b.) (3 points) Find a basis for the kernel of L . Show all work.

$$L(at^2 + bt + c) = 0$$

$$(a+b)t + (a-c) = 0$$

$$a+b=0$$

$$a-c=0$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 \end{array} \right] \xrightarrow{-r_1 \rightarrow r_2} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 \end{array} \right]$$

$$\downarrow r_2 + r_1 \rightarrow r_1$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & -1 & -1 & 0 \end{array} \right] \downarrow -r_2 \rightarrow r_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] \begin{array}{l} a-c=0 \\ b+c=0 \end{array}$$

(Problem 8 continued) More space for part (b), if needed:

$$\begin{aligned} c=r: \quad a &= r \\ b &= -r \\ c &= r \end{aligned}$$

$$rt^2 - rt + r = r(t^2 - t + 1)$$

$$\text{Basis: } \{t^2 - t + 1\}$$

c.) (3 points) Is $\mathbf{v} = 3t - 1$ in the range of L (also known as the image of P_2 under L)? Fully justify your answer.

$$L(at^2 + bt + c) = (a+b)t + (a-c) = 3t - 1$$

$$a+b=3$$

$$a-c=-1$$

$$\begin{bmatrix} 1 & 1 & 0 & | & 3 \\ 1 & 0 & -1 & | & -1 \end{bmatrix} \xrightarrow{\substack{-r_1 \rightarrow r_2 \\ \frac{1}{2}r_2}} \begin{bmatrix} 1 & 1 & 0 & | & 3 \\ 0 & -1 & -1 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 & | & 3 \\ 0 & 1 & -1 & | & -4 \end{bmatrix} \xrightarrow{-r_2 \rightarrow r_1} \begin{bmatrix} 1 & 0 & -1 & | & -1 \\ 0 & 1 & -1 & | & -4 \end{bmatrix} \quad \begin{aligned} b+c &= 4 \\ a-c &= -1 \\ b+c &= 1. \end{aligned}$$

$$b=3$$

$$a=0$$

Yes,

$$L(0t^2 + 3t + 1) = (0+3)t + (0-1)$$

$$= 3t - 1.$$

9. (4 points) Consider the vector space P_2 , and let $S = (t^2 + 2, t - 1, 3t + 4)$, $T = (2t^2, 3t, t + 1)$ be two ordered bases for P_2 . If $\mathbf{v}$ is a vector in P_2 with $[\mathbf{v}]_S = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$, find $[\mathbf{v}]_T$.

$$[\mathbf{v}]_S = 2(t^2 + 2) + 2(t - 1) + 0$$

$$= 2t^2 + 2t + 2$$

$$\begin{aligned} 2t^2 + 2t + 2 &= a(2t^2) + b(3t) + c(t + 1) \\ &= 2at^2 + (3b + c)t + c \end{aligned}$$

$$2a = 2 \rightarrow a = 1$$

$$3b + c = 2 \rightarrow 3b = 2 - 2 = 0 \rightarrow b = 0.$$

$$c = 2 \rightarrow c = 2$$

$$[\mathbf{v}]_T = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

Scratch paper page!!